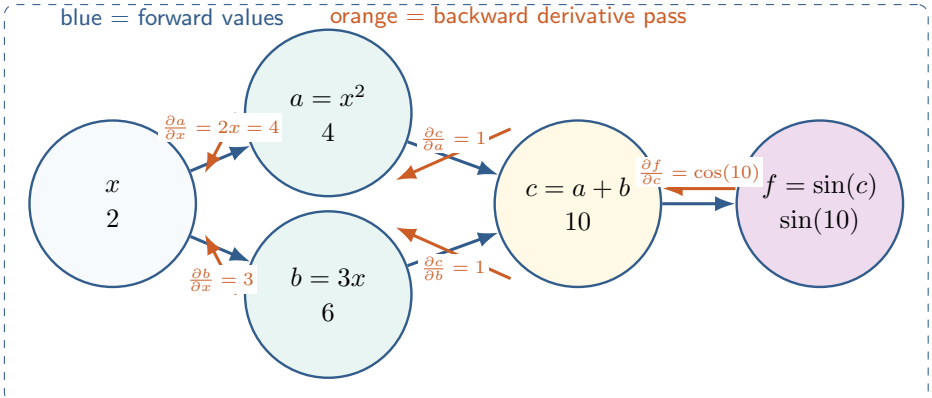
Autograd as a Worked Computational Graph

$f(x) = \sin(x^2 + 3x)$ at $x = 2$



Backward seed
Start with $\partial f / \partial f = 1$. Reverse pass: multiply local derivatives.

Accumulated gradient at the input

$$\frac{\partial f}{\partial x} = \cos(10)\big(1 \cdot 4 + 1 \cdot 3\big)$$
$$= \cos(10)(4 + 3).$$

PINN connection
For a neural option price $\widehat{p}(x, \theta)$, PyTorch builds a larger graph. Autograd walks it to obtain $\widehat{p}_x$, $\widehat{p}_\theta$, and then $\widehat{p}_{xx}$.